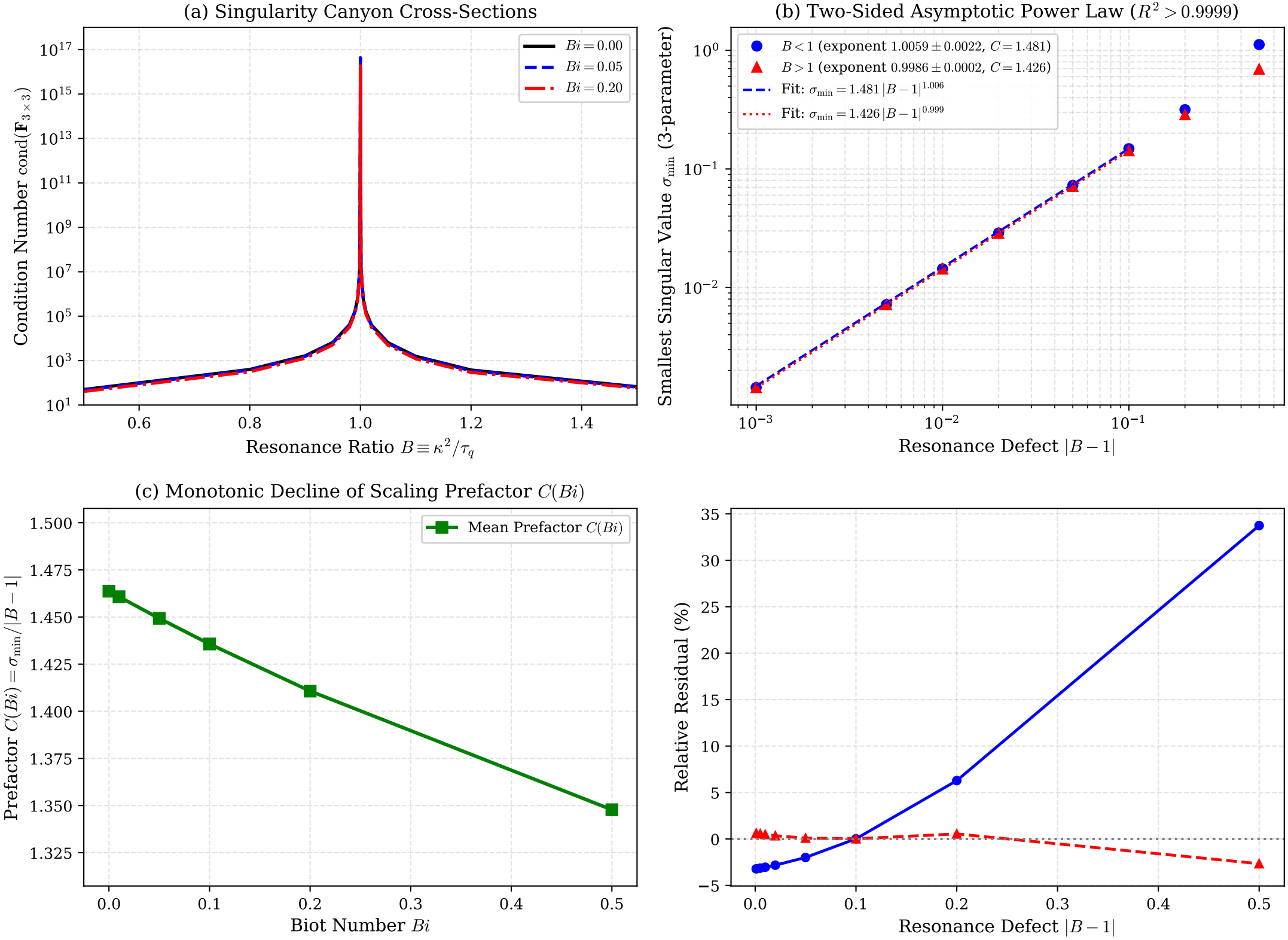
(a) Singularity Canyon Cross-Sections
Condition Number $\mathrm{cond}(\mathbf{F}_{3\times3})$
Resonance Ratio $B \equiv \kappa^2/\tau_q$
$Bi = 0.00$
$Bi = 0.05$
$Bi = 0.20$
(b) Two-Sided Asymptotic Power Law ($R^2 > 0.9999$)
Smallest Singular Value $\sigma_{\min}$ (3-parameter)
Resonance Defect $|B-1|$
$B<1$ (exponent 1.0059 ± 0.0022, $C = 1.481$)
$B>1$ (exponent 0.9986 ± 0.0002, $C = 1.426$)
Fit: $\sigma_{\min} = 1.481\,|B-1|^{1.006}$
Fit: $\sigma_{\min} = 1.426\,|B-1|^{0.999}$
(c) Monotonic Decline of Scaling Prefactor $C(Bi)$
Prefactor $C(Bi) = \sigma_{\min}/|B-1|$
Biot Number Bi
Mean Prefactor $C(Bi)$
Relative Residual (%)
Resonance Defect $|B-1|$